Traffic Sign Recognition Project Proposal

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link to Google Colab

Introduction

The aim of this project is to develop a deep learning system for Traffic Sign Recognition, capable of identifying traffic signs from images and providing their meanings to users. As an educational tool, it can help children build basic knowledge of traffic signs, allowing them to explore and learn about signs while walking or riding in a car. The system can also assist drivers who encounter unfamiliar signs, particularly during self-guided tours in foreign countries, thereby reducing potential safety risks. Furthermore, it can support individuals preparing for driving tests, such as the G1, by enabling them to study anytime and anywhere. In addition, the system can serve as an in-car guide by detecting signs such as speed limits, sharp curves, or school zones, which is especially useful on highways or when signs are partially obscured by foliage.

The project focuses on recognizing multiple images using multi-layered neural networks trained on appropriately sized datasets over several epochs. This enables the system to learn complex patterns and achieve fast, accurate recognition of a wide variety of traffic signs.

Illustration

Click here for clearer flow chart shown in Figure 1.

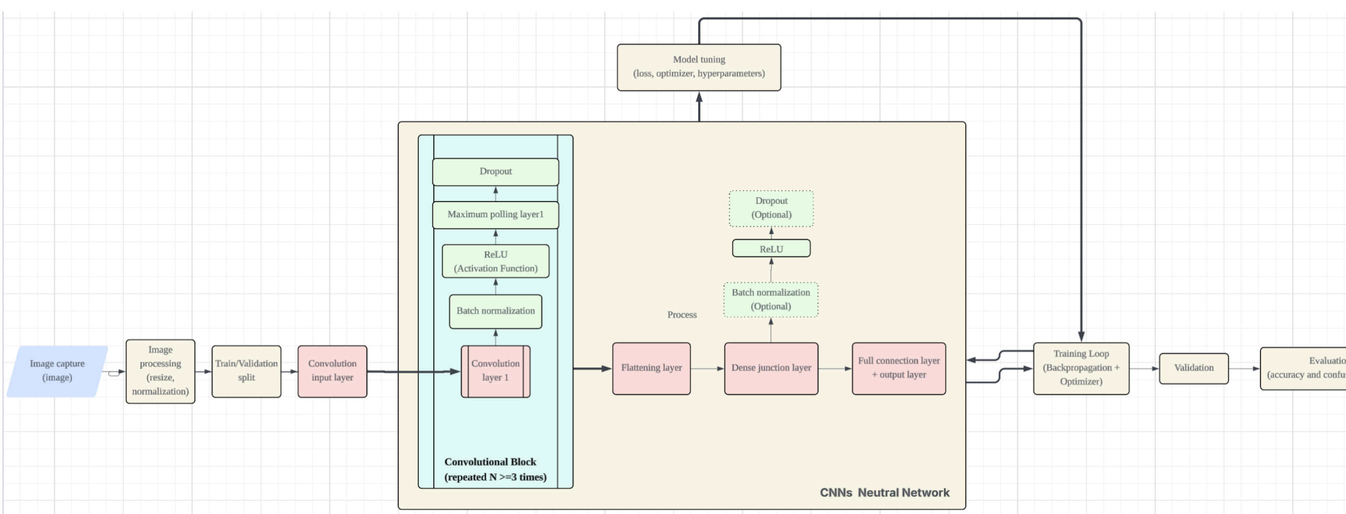

Figure 1: Flow chart illustrating overall CNN-based model used in project 'Traffic Sign Recognition'.

Background & Related Work

Traffic Sign Recognition (TSR) has been widely studied in computer vision and autonomous driving systems. The **German Traffic Sign Recognition Benchmarke** (GTSRB) Stallkamp et al. (2012), containing over 50,000 high-quality images across more than 40 classes, is commonly used to train classification models and captures real-world variations such as angle, lighting, and partial occlusion. Mobile applications like **Mapillary** Traffic Sign Recognition leverage CNNs and image processing techniques (e.g., HSV segmentation) to detect traffic signs in real time, providing educational support and driver assistance Qiao (2023). TSR is also critical in **autonomous vehicles**, where systems like Tesla and Waymo integrate cameras with deep learning models for fast, precise sign detection to enhance driving safety Alawaji et al. (2024). Research has shown that **CNNs** can automatically learn features from traffic sign images, perform better than traditional methods and achieving high classification accuracy Mukovoz et al. (2024). For more complex scenarios, the **German Traffic Sign Detection Benchmark** (GTSDB) **?** provides full road images with cluttered backgrounds and occlusions, supporting robust detection model training.

Data Processing

The primary dataset used in this project is the German Traffic Sign Recognition Benchmark (GTSRB) Real-Time Computer Vision Group, Institut für Neuroinformatik, Ruhr-Universität Bochum (2019), which contains over 50,000 labeled traffic sign images across 43 classes for classification tasks. To expose the model to more complex road scenes, the German Traffic Sign Detection Benchmark (GTSDB) **?** is also used, which includes full road images with traffic signs under challenging conditions such as occlusion and background clutter.

Images from both datasets are cleaned by removing corrupted or low-quality samples. All images are resized to a fixed resolution (e.g., 32×32 pixels for classification) and normalized to the range [0,1]. For GTSDB images, bounding boxes are used to crop traffic signs before classification. Labels are encoded numerically.

Data augmentation techniques such as random rotation, brightness adjustment, and translation are applied to improve robustness. A fixed random seed ensures that the dataset is split the same way every time, allowing experiments and results to be reproduced consistently..

Architecture

This project uses Convolutional Neural Networks (CNNs) to handle the high-dimensional, complex images of traffic signs, which vary in size, lighting, and angle. CNNs automatically extract features like edges and shapes, and transfer learning with pretrained models enables high accuracy even with limited data, ensuring robust recognition across diverse signs and conditions.

Baseline Model

As a baseline, a traditional machine learning approach is used instead of deep learning. All traffic sign images are resized to a fixed resolution and converted to grayscale. Histogram of Oriented Gradients (HOG) features are extracted to capture basic edge and shape information. These features are then classified using a linear Support Vector Machine (SVM). Instead of relying on handcrafted features, CNNs can automatically learn features from images, which brings more efficient training.

Ethical Considerations

This Traffic Sign Recognition system raises ethical concerns related to privacy, misuse, and safety. Uploaded images may unintentionally reveal sensitive information such as user location, surroundings, or identifiable individuals, requiring careful and ethical data collection. The system could also be misused during official driving tests (e.g., the G1), where external assistance should be restricted. Additionally, the model has limitations due to the black-box nature of deep neural networks and potential biases or gaps in the training data, which may lead to misclassification under rare conditions.

Link to Github or Colab Notebook

Google Colab

GitHub

References

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